AI.207

Mobile Robotics

COURSE OVERVIEW

Robots that move: locomotion, wheeled and legged platforms, navigation, and mobile manipulation.



ILLUSTRATIVE HERO IMAGERY

1

SESSION

60 MINUTES

ORIENTATION: ROBOTS THAT MOVE

KEY TOPICS

- Locomotion types
- Wheeled vs. legged vs. aerial
- Mobile robot applications
- Challenges of mobility

TEACHING FOCUS

Survey how robots get from place to place.

KEY TAKEAWAY Mobility expands a robot's reach.

2

SESSION

60 MINUTES

WHEELED LOCOMOTION

KEY TOPICS

- Differential drive
- Ackermann steering
- Omnidirectional and mecanum
- Odometry

TEACHING FOCUS

Model and control wheeled motion.

KEY TAKEAWAY Wheels are efficient on structured floors.

MOBILE ROBOTICS

3

SESSION

60 MINUTES

LEGGED LOCOMOTION

KEY TOPICS

- Bipedal, quadruped, hexapod
- Gaits and walking
- Balance and ZMP
- Boston-dynamics-style platforms

TEACHING FOCUS

Achieve stable walking on legs.

KEY TAKEAWAY Legs go where wheels cannot.

4

SESSION

60 MINUTES

MOBILE NAVIGATION

KEY TOPICS

- Waypoint following
- Obstacle avoidance
- Potential fields
- Vector field histograms

TEACHING FOCUS

Move safely toward a goal.

KEY TAKEAWAY Navigation is motion with awareness.

5

SESSION

60 MINUTES

MANIPULATION ON THE MOVE

KEY TOPICS

- Mobile bases with arms
- Coordination of base and manipulator
- Whole-body control
- Case studies

TEACHING FOCUS

Combine movement with manipulation.

KEY TAKEAWAY Mobile manipulators do work across spaces.

6

SESSION

60 MINUTES

MOBILE ROBOTICS & REVIEW

KEY TOPICS

- Platform demo and assessment
- Course recap
- Toward AI for robotics (AI.208)
- Field trial

TEACHING FOCUS

Operate a mobile robot end-to-end.

KEY TAKEAWAY A mobile robot must go, sense, and act.